IT 213: Networking

Lab 10: The Control Plane – Local Routes, BGP, and Link-State Databases

Concordia University of Edmonton

Objective

In Chapter 5, we explored how the Control Plane builds forwarding tables using intra-AS protocols (OSPF) and inter-AS protocols (BGP). Since we cannot easily log into enterprise backbone routers, this lab will teach you how to inspect your own computer's local routing table, how to use public "Looking Glass" servers to view actual BGP routing tables from major ISPs, and how to mathematically model a Link-State database.

Part 1: Inspecting the Local Forwarding Table

Even your local computer runs a simplified version of a forwarding table to know where to send packets (e.g., to the local network or to the default gateway).

1. Open your Command Prompt (Windows) or Terminal (Mac/Linux).
2. Execute the following command to view your routing table:
   - **Windows:** `route print` (Look at the "IPv4 Route Table" section)
   - **Mac/Linux:** `netstat -rn`
3. **Lab Report Question 1:** Identify the "Network Destination" of `0.0.0.0` (or `default`). What is the "Gateway" IP address listed for this destination? Why does all traffic destined for the broader Internet get sent to this specific Gateway address?
4. **Lab Report Question 2:** Look at the "Interface" column. What is the IP address of your specific machine on this local network?

Part 2: BGP and the AS-PATH (Looking Glass)

BGP routes traffic between Autonomous Systems (AS) using attributes like the AS-PATH. Network engineers use public "Looking Glass" servers to troubleshoot BGP routes across the global Internet.

1. Open your web browser and go to a public BGP Looking Glass (e.g., `https://lookingglass.centurylink.com/` or `https://lg.he.net/`).
2. Select a router location (e.g., New York, London, Tokyo).
3. Choose the "BGP Route" or "BGP" command option.

4. Enter a target IP address. (You can use `8.8.8.8` for Google's DNS, or find Concordia University of Edmonton's public IP).

5. Execute the query.

6. **Lab Report Question 3:** Look at the output results. Identify the `AS-PATH` (usually a string of numbers like `3356 15169`). How many distinct Autonomous Systems does this specific BGP route pass through to reach the destination?

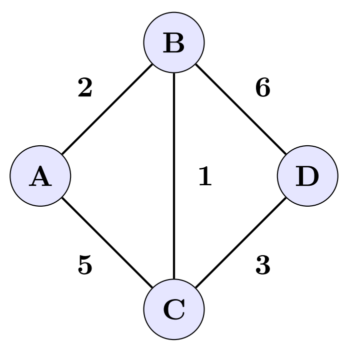
7. **Lab Report Question 4:** According to your textbook (Section 5.4), how do BGP routers use this AS-PATH attribute to prevent routing loops?

Part 3: Building the Link-State Database

In our lecture on the Control Plane, we discussed that before an OSPF router can run Dijkstra's algorithm, it must first build a complete mathematical map of the network. It does this by compiling the Link-State broadcasts from other routers into an **Adjacency Matrix** (or Cost Matrix).

Example Topology

Imagine a small Autonomous System with 4 routers (A, B, C, D) and the physical link costs shown in the graph below:



If Router A is building its database, it maps these costs into a 2D matrix.

- If a direct physical link exists, the matrix stores the cost.
- If a node is referencing itself on the main diagonal, the cost is 0.
- If no *direct* physical link exists between two nodes (e.g., A to D), the cost is set to Infinity (∞).

The resulting Adjacency Matrix for this network is:

$$\begin{bmatrix} 0 & 2 & 5 & \infty \\ 2 & 0 & 1 & 6 \\ 5 & 1 & 0 & 3 \\ \infty & 6 & 3 & 0 \end{bmatrix}$$

Note: You will need your class notes to recall exactly how Dijkstra's algorithm initializes and processes this matrix step-by-step for your upcoming quiz.